

DANIELE BORTOLUZZI

Software and IoT Architect / R&D

CONTACT

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SKILLS

Java
Spring Framework
Quarkus
Scala
Kubernetes
AWS
Terraform
Linux
Python
TypeScript
C++
ROS2

Teamwork
Problem-solving
Mentoring
Leadership
Innovation
Decision-making
Communication

PROFILE

15 years of experience from design through production: cloud-native backends, data and AI in production, AWS/Terraform—with team leadership, turnkey delivery, and mentoring.

I **lecture** on cloud and distributed systems—containers, microservices, Spring/Spring Cloud, Docker, Kubernetes, and OpenStack—for universities and partner programmes. One current engagement is the University of Turin's Master's in Cloud Computing (COREP); I am open to guest lectures, modules, and corporate training at other institutions as well.

Alongside client work, I keep an active **research** thread in swarm intelligence and IoT, with *rovers*, *drones*, and embedded prototypes tied to experiments and field deployments.

WORK EXPERIENCE

Freelance IT Consultant **03/2025 – now**
Remote, Turin

Independent consultant: software architecture, turnkey delivery, team leadership, and mentoring for clients building cloud-native platforms, data products, and AI-driven industrial solutions (including Agriculture 4.0).

- Microservice architectures (independent services with Quarkus), React front ends consuming those APIs, and integration across distributed components.
- Data ingestion pipelines and dashboards for monitoring, exploration, and operational use of large datasets.
- Deployment on AWS and infrastructure-as-code: provisioning environments and machines with Terraform.
- AI projects for data analysis and computer vision applied to Agriculture 4.0 use cases.

Technologies include:

- Quarkus, React (TypeScript), microservices, Redis, AWS, Terraform, Python (AI and data).

Achievements include:

- Acting as team leader, software architect, and hands-on engineer on turnkey engagements.
- Mentoring developers and customer teams through design, implementation, and production rollout.

EDUCATION

09/2013 – 07/2022

Master's Degree in Computer Science (Level 7 EQF)
University of Turin

Graduation grade: *110/110 with honors and dignity of publication*.

09/2007 – 06/2011

Bachelor's Degree in Computer Science (Level 6 EQF)
Ca' Foscari University of Venice

Graduation grade: *104/110*.

09/2002 – 06/2007

Scientific High School Diploma
Liceo Scientifico Statale "G. Marconi" di Conegliano (TV)
National Plan for Informatics.
Final grade: 90/100.

PUBLICATIONS

G. Audrito, A. Basso, D. Bortoluzzi, F. Damiani, G. Scarso, G. Torta.
"Exploiting Aggregate Programming in a Multi-Robot Service Prototype",
In: Proc. 17th Workshop on Programming Language Approaches to Concurrency and Communication-cEntric Software (PLACES 2026). *Electronic Proceedings in Theoretical Computer Science* (EPTCS), vol. 444, 2026, pp. 45–57. arXiv:2604.06876

Contract Lecturer **11/2025 – 04/2026**
COREP (Turin)

Teaching on site for the Master's Program in Cloud Computing for Artificial Intelligence and Internet of Things, a first-level university master's degree of the University of Turin, organized with COREP.

- Led the course "Containerization and Virtualization Platforms".
- Covered Spring Boot, microservices, Docker, Kubernetes, and OpenStack.

Technologies include:

- Spring Boot, microservices, Docker, Kubernetes, OpenStack, Java.

Research Fellow **03/2023 – 02/2025**
University of Turin, Department of Computer Science

Within the MoVeRe research group at the University of Turin, I conduct research activities related to the Aggregate Programming paradigm, which allows leveraging coordination between heterogeneous entities.

- Participation in the PRIN 2020 project named COMMON-WEARS.
- Collaboration in the organization and development of verticalizations and PoC in the Smart Mobility and Industry 4.0 sectors.
- Teaching support activities for the courses "Web Technologies" and "Programming for mobile devices".

Technologies include:

- ROS2 (Python, C++), Gazebo (Python), Zephyr (C, C++), Arduino (C), Spring Boot (Java), Angular (TypeScript).

Achievements include:

- A new platform for IoT and robotics using Aggregate Programming and ROS2.
- Working on PRIN and European projects, using R&D technology in specialized industry.
- Co-authored peer-reviewed and conference publications on aggregate programming, coordination, and IoT/robotics (see Publications).

Solution Architect **10/2021 – 03/2023**
GFT Italia S.r.l. (Turin)

Consulting activities in software architectural analysis and design of microservice-based solutions for insurance market clients.

- Technical reference and coordination of development teams.
- Technical training and coaching activities for resources.
- Design and coordination of the IT solution for the new core customer registry system for a lead insurance group.

Technologies include:

- Java 11+, Spring Boot, Oracle, Spring Cloud, Redis, DB2, MyBatis, Tomcat, IBM WebSphere, SQL, Gitlab, Maven, REST, IBM MQ, Apache Kafka.

G. Audrito, D. Bortoluzzi, F. Damiani, G. Scarso, G. Torta, A. Basso, M. Cochi, L. Gusman, L. Comba, P. Gay, P. Dal Zovo, G. Galati, F. Gallo, A. Grdadolnik, M. Pescarollo, P. Pisano.
"Lambdas at the Far Edge: a Tale of Flying Lambdas and Lambdas on Wheels",
In: Logics and Type Theory (LTT 2026): essays dedicated to Stefano Berardi. *Electronic Proceedings in Theoretical Computer Science* (EPTCS), vol. 441, 2026, pp. 19–45. arXiv:2603.04008

Audrito, G., Bortoluzzi, D., Damiani, F., Scarso, G., Torta, G.
"An Enhanced Exchange Operator for XC." In: Castellani, I., Tiezzi, F. (eds) Coordination Models and Languages. COORDINATION 2024. Lecture Notes in Computer Science, vol 14676. Springer, Cham

L. Nitti, D. Bortoluzzi, A. Biglia, D. Ricauda Aimomino, G. Torta, G. Audrito, F. Damiani, P. Gay. A. Rapp, L. Comba.
"UAS-swarm based surveillance system for UGVs safety functionality", Proc. of the 15th European Conference on Precision Agriculture, 2024, Barcelona (Spain)

A. Basso, D. Bortoluzzi and G. Torta.
"Implementation of an IoT Wearable Prototype on a Standard AI Architecture," 2022 IEEE Intl Conf on Dependable, Autonomic and Secure Computing, Intl Conf on Pervasive Intelligence and Computing, Intl Conf on Cloud and Big Data Computing, Intl Conf on Cyber Science and Technology Congress (DASC / PiCom / CBDCom / CyberSciTech), Falerna, Italy, 2022, pp. 1-5

A. Basso, D. Bortoluzzi, G.Torta, D. Pau, L. Chariglione, F. Damiani.
"Architecture standardization for AI deployment on tiny microcontrollers",
Proc. of the 12th IEEE Int. Conf. on Consumer Technology, 2-6 September 2022, Berlin (Germany)

Achievements include:

- Managed and detailed complex systems for enterprise-level projects.
- Led a team of 20+ members, with different roles and skill sets.

Senior Software Developer / Technical Leader **01/2017 – 10/2021**
Xeffe S.r.l. (Turin)

Analysis, coordination, and development of Enterprise Solutions for the finance market.

- Use of Domain Driven Design and Enterprise Integration Patterns.
- Development and maintenance of projects for finance market.
- Development and maintenance of proprietary frameworks based on Apache Camel and Play Framework.

Technologies include:

- Java 8+, Scala, JavaScript, Spring, SQL, Apache Camel, Play Framework, Oracle, H2, MyBatis, GitHub, Ansible, Maven, nginx, SOAP, REST, jQuery, BeanIO, Quartz.

Achievements include:

- High customer satisfaction in the design and implementation of customer-specific projects using proprietary frameworks.

Software Architect / Full Stack Developer **06/2015 – 01/2017**
Audacia S.r.l. (Turin)

Coordination and development of an Intelligent Big Data Analysis system to extract significant information from heterogeneous data sources.

- Analysis and development of software solutions for insurance companies.
- Maintenance of Java applications for finance market.

Technologies include:

- Node.js, Google Cloud Platform, JavaScript, ArangoDB, Java, SQL, AQL, Loopback, Express, AngularJS, Oracle, GitHub, GIT, Grunt, Bower, NPM, Bootstrap.

Achievements include:

- Implemented a draft solution for Intelligent Big Data Analysis using cutting-edge technologies.

Software Engineer / IT Manager **11/2014 – 06/2015**
Creatiweb S.r.l. (Turin)

Full-stack development of a proprietary booking engine for campsites and villages.

- Development of WebServices to integrate the booking engine with accounting and channel manager software.
- Implementation of new features of the booking engine application.
- Performance analysis and tuning of company databases.

- Manager of the server farm.

Technologies include:

- Java, SQL, Linux, JavaScript, PHP, XSD, XML, CSS, HTML, SQL, Bash Script, Firebird DBMS, Tomcat, jQuery, XPath, XMLBeans, SOAP.

Achievements include:

- Served as the main point of contact for the IT department, supporting both internal and external clients.

Software Engineer / Developer **11/2011 – 11/2014**
NaturalBooking S.r.l. (Turin)

Development of the back-end of Hospes, a booking engine used by many campsites and villages in Italy.

- Development of SOAP WebServices to integrate the Booking Engine with accounting and channel manager software.
- IT administration of servers hosting the application.

Technologies include:

- Java, Firebird DBMS, JavaScript, SQL, XML, CSS, HTML, Bash Script, Tomcat, SOAP.

Analyst / Developer **06/2011 – 11/2011**
Prosa S.r.l. (Venice)

Implementation of server-side and client-side features of the EmotiVending system, a vending machine equipped with a touch multimedia interface.

- Analysis and development of the new VeBox system, a control center for vending machine controllers, in Java.
- Development of an application for testing hardware of vending machines.

Technologies include:

- C++, Ruby, Java, Linux, Tomcat, Java Swing, Restlet.

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